- We know both xy and $x+y$ are even.
- By Defn. 1, $\exists$ integers K_3 and K_4 s.t. $xy = 2K_3$ and $(x+y) = 2K_4$.

(Note: We shall use this information in Case 3)

$(x+y) - x = 2K_4 - (2K_1+1) = \boxed{2(K_4 - K_1 - 1) + 1}$

It's easy to see that $(K_4 - K_1 - 1)$ is an integer. It's easy to see that from Defn. 1, $(x+y) - x$ is odd.

$(x+y) - x = y$, which is even. $\therefore (x+y) - x$ is even.

From Defn. 1, an integer cannot be both even and odd at the same time.

$\therefore$ This is a contradiction.

So, ~~x cannot be even~~ x is even and y is odd and vice-versa is a False case.

(3) $\neg p$ is True, $\neg s$ is True. $\therefore$ Both x and y are odd.

By Defn. 1, $\exists$ integers K_1 and K_2 s.t. $x = 2K_1 + 1$ and $y = 2K_2 + 1$.

$y = \frac{xy}{x}$. $\boxed{\therefore \frac{xy}{x} \text{ is an odd integer}}$

$\frac{xy}{x} = \frac{2K_3}{2K_1+1} = 2\left(\frac{K_3}{2K_1+1}\right)$. If $2K_1+1 \nmid K_3$, then $\frac{xy}{x}$ is not an integer.

Else, From Defn. 1, $\boxed{\frac{xy}{x} \text{ is an even integer}}$

From Defn. 1, $\frac{xy}{x}$ can't be both even and odd at the same time.

Obviously, $\frac{xy}{x}$ can't be an integer and also not an integer at the same time. $\therefore$ This is a contradiction. So the case where both x and y are odd is a False case.

So, $(\neg p \vee \neg s)$ is False. By De Morgan's Law, $\neg(p \wedge s)$ is False.

By defn. of AND, $\boxed{(p \wedge q) \wedge \neg(p \wedge s) \text{ is False.}}$